

# Complete Java Notes

**Introduction:** Java is a high-level, object-oriented programming language developed by Sun Microsystems (now Oracle). It follows 'Write Once, Run Anywhere' using JVM.

## Features of Java:

- Object-Oriented
- Platform Independent
- Secure
- Robust
- Multithreading
- High Performance

## Java Architecture:

Java code → compiled by javac → bytecode → executed by JVM → runs on any platform.

## OOP Concepts:

- Class & Object
- Inheritance
- Polymorphism
- Encapsulation
- Abstraction

## JVM, JRE, JDK:

JVM: Executes bytecode

JRE: Runtime environment

JDK: Development kit

## Example Program:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println('Hello World');  
    }  
}
```

## Applications of Java:

- Web Development
- Android Apps
- Desktop Apps
- Enterprise Systems

**Conclusion:** Java is widely used due to its portability, scalability, and strong ecosystem.